

Oguzhan Yetimoglu

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SUMMARY

Software engineer with 3+ years across full stack, backend, and cloud native development, and an M.Sc. in Informatics from the Technical University of Munich. I build systems end to end: Python and TypeScript backends, React and Next.js frontends, and containerized deployments with Docker and Kubernetes on AWS, GCP, and Azure. I also ship LLM and ML driven features, and I work well close to the product in small teams.

WORK EXPERIENCE

Siemens (Munich) - Software Engineer (Working Student)

Jun 2024 - Sep 2025

- Reworked Apache NiFi dataflows on a cloud based security analytics platform (Java, Azure), clearing pipeline bottlenecks and fixing inter-service communication to improve ingestion and monitoring reliability.
- Built an LLM powered network intrusion detection module (Python, anomaly detection) that strengthened detection quality in distributed data pipelines.
- Made the data pipeline configurable through JSON inputs, letting non-technical users build and adjust pipelines without writing code.
- Optimized microservice Docker builds, cutting image size and improving build reproducibility by updating base images, dependencies, and build scripts.

Ludwig Maximilian University (Munich) - Student Research Assistant

Oct 2023 - Jun 2024

- Processed large scale brain MRI neuroimaging datasets using ML and statistical modeling techniques to classify disease presence and subtype patterns.
- Parallelized preprocessing and processing pipelines on HPC clusters, which sharply cut feature extraction and model execution time.
- Improved research workflow reliability by automating ETL steps, standardizing feature extraction procedures, and optimizing job scheduling to reduce idle compute cycles.

Aimyca GmbH (Munich) - Software Engineer (Working Student)

Mar 2022 - Mar 2023

- Delivered full stack features using Python, Node.js, React, Vue, TypeScript, and REST APIs in an agile SaaS environment.
- Built reusable React and Vue components and improved the product's UI/UX by reworking interaction flows and refining the interface.
- Built CI/CD pipelines with Bitbucket Pipelines on GCP to stabilize deployments and improve system reliability.
- Enhanced conversational AI features through prompt engineering and refined extraction logic.
- Worked closely with the founding team in a small engineering setup, contributing to technical discussions and product decisions.

NETLAB, Bogazici University - Graduate Research Assistant / Software Engineer

Feb 2020 - Dec 2021

- Implemented simulation algorithms and C++/Python testbeds for TUBITAK funded molecular communication research, improving experiment reproducibility.
- Built embedded data acquisition integrations on Raspberry Pi and Arduino for sensor and actuator based experimental testbeds.

EDUCATION

Technical University of Munich - M.Sc. Informatics (Grade: 1.6, 1.0 = best)

Sep 2021 - Sep 2025

Bogazici University - M.Sc. Computer Engineering (GPA: 4.0, 4.0 = best)

Sep 2019 - Sep 2022

Bogazici University - B.Sc. Computer Engineering

Sep 2014 - Jul 2019

PROJECTS

LoRemaster - Mobile Quiz App for Fantasy & Sci-Fi Fandoms (2026)

- Architected a cross-platform mobile quiz app spanning different fictional and mythological universes (LOTR, Dune, Greek mythology, Lovecraft) with daily curated quizzes and an endless mode.
- Built a TypeScript monorepo spanning four apps (mobile, admin, marketing, and a Firebase Cloud Functions backend) with shared code, using React Native (Expo), Next.js, Zustand, and Firestore for scheduled jobs, event-driven triggers, and server-side validation.

- Engineered an LLM-powered question generation pipeline with LiteLLM, orchestrating multi-provider models through prompt engineering, automated validation, and provider fallback for reliability and cost optimization.
- Set up CI/CD with GitHub Actions across the monorepo, covering linting, type checks, Jest unit tests, and Maestro end-to-end mobile tests, with EAS for Android builds.
- Designed data access and infrastructure for cost efficiency, minimizing Firestore reads and writes and keeping overall resource usage lean.

Elysio.ai - AI Driven Virtual Fashion Prototype (2024)

- Built a full stack generative AI prototype using Python (FastAPI), TypeScript, React (Next.js), and MUI.
- Integrated Stable Diffusion and Replicate API for personalized image generation, pose controlled synthesis, and NeRF style 3D reconstruction.
- Designed modular cloud deployment on AWS using Docker, CI/CD, and environment separation for scalable inference workloads.
- Collaborated with the business concept team to translate product requirements into technical generative AI workflows.

Algorithmic Textual Comparison Platform (TUM IDP, 2024)

- Developed a full stack platform for financial report comparison with NLP based semantic similarity (TF IDF, BERT embeddings, cosine similarity).
- Built the frontend using Vue.js, Nuxt.js, and Tailwind CSS with dynamic interfaces for upload, search, and comparison visualization.
- Implemented backend APIs in Python (Django) with PostgreSQL, authentication, and document management.
- Designed the core text comparison pipeline and coordinated task planning and development sprints.

SKILLS

Programming and Frameworks	Python, TypeScript, JavaScript, Java, C++, React, React Native, Expo, Next.js, Vue, Nuxt.js, Node.js, FastAPI, RESTful APIs
Cloud and DevOps	Docker, Kubernetes, CI/CD, GitHub Actions, GitLab CI, Linux, AWS, GCP, Azure, container orchestration, microservices, cloud native environments
Data and Machine Learning	Pandas, NumPy, SciPy, scikit-learn, ML, NLP, LLMs, embeddings, vector databases, semantic search, data pipelines
Databases and Storage	PostgreSQL, MongoDB, Firebase (Firestore)
Tools and Engineering Practices	OOP, debugging, testing, API design, performance optimization, monitoring, observability, logging pipelines, automated deployments, Bash, shell scripting, AI-assisted development
Frontend Technologies	Tailwind CSS, Material UI, component based architecture, responsive UI design

PUBLICATIONS

1 journal and 3 conference papers in signal processing and molecular communication.

LANGUAGES

Turkish (Native) • English (C1) • German (A2)